

# Encapsulation

Is a mechanism of wrapping attributes and methods into objects—an object's attributes and methods are intimately related. Objects may communicate with one another, but they're normally not allowed to know how other objects are implemented. This **information hiding** is important to good software engineering.

In encapsulation, variables of class will be hidden from other classes, and can be only accessed through the methods of their current class. The declaring of instance variables with access modifier private is known as **data hiding** or information hiding.

## Example,

```
Bank.java X BankAccount.java
1
2 public class Bank {
3     public static void main(String[] args) {
4         BankAccount customer=new BankAccount();
5         customer.balance=100000;
6         customer.setBalance(1000000);
7         System.out.print("Balance Information :"+ customer.getBalance());
8     }
9 }
```

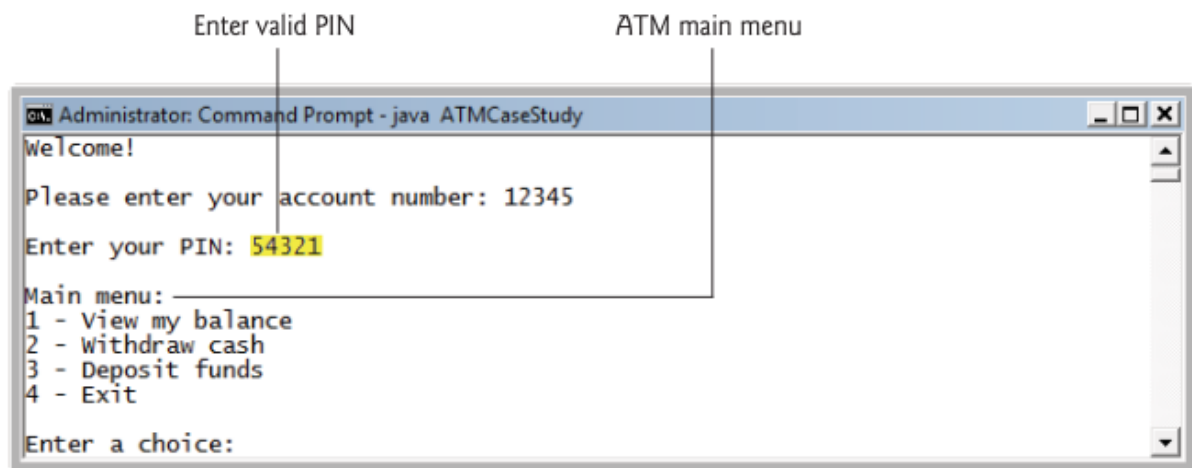
```
Bank.java BankAccount.java X
1 public class BankAccount {
2     private double balance;
3
4     public BankAccount(){
5         balance=100000;
6     }
7
8     double getBalance(){
9         return balance;
10    }
11
12    void setBalance(double ammount){
13        //WE CAN SET ANY REQUIRED CONDITIONS
14        balance+=ammount;
15    }
16
17 }
```

## Why we need access control?

- 1) Protect private information
- 2) Clarify how others should use your class
- 3) Keep implementation separate from interface

## Homework

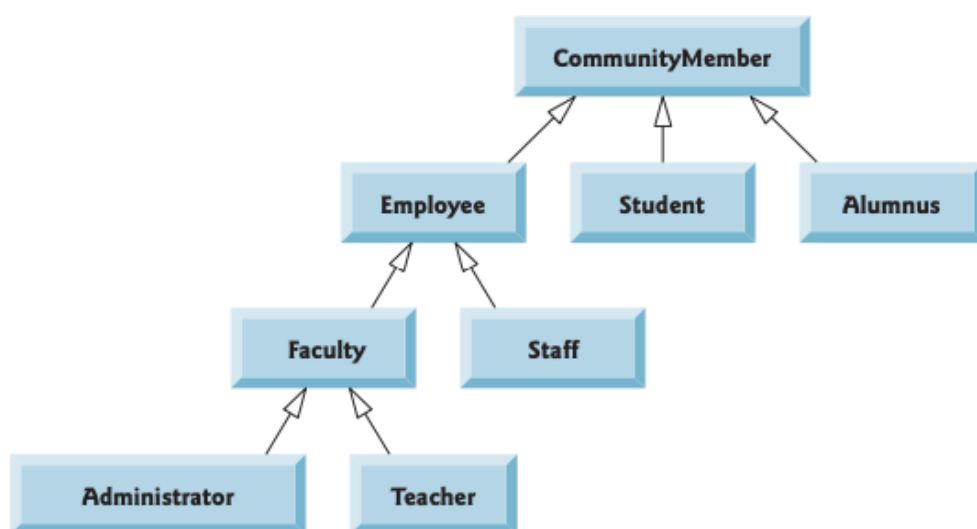
Design the following application:



## Inheritance

is a form of software reuse in which a new class is created by absorbing an existing class's members and embellishing them with new or modified capabilities. With inheritance, you can save time.

When creating a class, rather than declaring completely new members, you can designate that the new class should inherit the members of an existing class. The existing class is called the **superclass**, and the new class is the **subclass**. Each subclass can become a superclass for future subclasses. For example, A university community has thousands of members, including employees, students, and alumni. Employees are either faculty or staff members. Faculty members are either administrators or teachers. students can be graduate or undergraduate students.



## Note

- Inheritance reduces program development time.
- The direct superclass of a subclass (specified by the keyword **extends** in the first line of a class declaration) is the superclass from which the subclass inherits.
- An indirect superclass of a subclass is two or more levels up the class hierarchy from that subclass.
- Java does not support multiple inheritance.
- The annotation **@Override** indicates that a method should override a superclass method.

## Example,

```
Registration.java X Student.java College.java
1 package inheritance;
2
3 public class Registration {
4     public static void main(String[] args) {
5         Student st=new Student();
6         st.firstname="Ali";
7         st.secondname="Kadhim";
8         st.cname="CS & Maths";
9
10        System.out.print(st.getInfo());
11    }
12 }
```

```
Registration.java Student.java X College.java
1 package inheritance;
2
3 public class Student extends College{
4     public String firstname;
5     public String secondname;
6
7     String getInfo() {
8         return (firstname + " " + secondname+ "\nStuding in : " + cname);
9     }
10
11 }
```

```
Registration.java Student.java College.java X
1 package inheritance;
2
3 public class College {
4     String cname;
5     private String cID;
6
7     String getInfo() {
8         return ("College Name :"+ cname);
9     }
10
11 }
```

## Homework

Using programming show the effect of the keyword **final** usage for instance variable.